Correlation Calculation — Line by Line

This guide explains how to calculate the correlation coefficient manually using your student dataset. We calculate ONE PAIR of columns at a time. Four columns are NOT multiplied together.

1. Your Dataset

Study_Hours	Attendance_Pct	Assignment_Score	Exam_Score
2	65	55	58
3	68	60	62
4	72	64	67
5	75	68	70
6	78	72	74
7	80	75	78
8	83	78	81
9	85	82	84
10	88	85	87
11	90	88	90
12	92	91	93
13	95	94	96

2. Correlation Formula

$r = \Sigma[(X – X\blacksquare)(Y – \blacksquare)] / \sqrt{}\{\Sigma(X – X\blacksquare)^2 \times \Sigma(Y – \blacksquare)^2\}$

Remember: choose TWO columns. X is the first column and Y is the second column.

3. Full Example: Study_Hours vs Exam_Score

We want to find the value 0.997212274. Take only Study_Hours and Exam_Score.

Step 1 — Find the mean of Study_Hours

Sum = 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10 + 11 + 12 + 13 = 90
Number of observations = 12
X■ = 90 / 12 = 7.5

Step 2 — Find the mean of Exam_Score

Sum = 58 + 62 + 67 + 70 + 74 + 78 + 81 + 84 + 87 + 90 + 93 + 96 = 940
Number of observations = 12
■ = 940 / 12 = 78.3333

Step 3 — Calculate deviations and products

X	Y	X–7.5	Y–78.3333	(X–X■)(Y–■)	(X–X■)²	(Y–■)²
2	58	-5.5	-20.3333	111.8333	30.25	413.44
3	62	-4.5	-16.3333	73.5000	20.25	266.78
4	67	-3.5	-11.3333	39.6667	12.25	128.44
5	70	-2.5	-8.3333	20.8333	6.25	69.44
6	74	-1.5	-4.3333	6.5000	2.25	18.78
7	78	-0.5	-0.3333	0.1667	0.25	0.11
8	81	0.5	2.6667	1.3333	0.25	7.11
9	84	1.5	5.6667	8.5000	2.25	32.11
10	87	2.5	8.6667	21.6667	6.25	75.11
11	90	3.5	11.6667	40.8333	12.25	136.11
12	93	4.5	14.6667	66.0000	20.25	215.11
13	96	5.5	17.6667	97.1667	30.25	312.11
TOTAL				488	143	1674.67

4. Step 4 — Add the Columns

After completing the table, take the totals you need:

$\Sigma(X-X\blacksquare)(Y-\blacksquare) = 488$
$\Sigma(X-X\blacksquare)^2 = 143$
$\Sigma(Y-\blacksquare)^2 = 1674.6667$

5. Step 5 — Put Values into the Formula

$r = 488 / \sqrt{(143 \times 1674.6667)}$

First multiply: $143 \times 1674.6667 = 239477.33$

Then square root: $\sqrt{239477.33} \approx 489.364$

Finally: $r = 488 / 489.364 \approx 0.997212$

Answer: r ≈ 0.997212274

6. How to Calculate All 4 Columns

Pair	X	Y	Result
1	Study_Hours	Attendance_Pct	0.9968213
2	Study_Hours	Assignment_Score	0.99750139
3	Study_Hours	Exam_Score	0.997212274
4	Attendance_Pct	Assignment_Score	0.999265256
5	Attendance_Pct	Exam_Score	0.999281638
6	Assignment_Score	Exam_Score	0.999379742

7. Important: You Never Multiply 4 Columns Together

There are 4 variables, so you calculate correlation between TWO variables at a time. For 4 variables, the number of unique pairs is $4 \times 3/2 = 6$. The diagonal values in the correlation matrix are 1 because every variable has a perfect correlation with itself.

8. Easy Memory Method

For every pair: (1) Find Mean X → (2) Find Mean Y → (3) X–Mean X → (4) Y–Mean Y → (5) Multiply deviations → (6) Square X deviation → (7) Square Y deviation → (8) Add columns → (9) Put totals into formula.

Interpretation: Values close to +1 mean a very strong positive relationship. For example, 0.9972 means Study Hours and Exam Score have an extremely strong positive linear relationship in this dataset.